

Design and Implementation of a Non-contact Thermometer that Records Body Temperature Individually using Forehead Wrinkle Image

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ABSTRACT

With the novel coronavirus disease (COVID-19) pandemic, there has been a significant increase in the opportunities to measure body temperature when using public facilities and to check one's daily health condition. Standing thermometer and handheld devices have been introduced due to their non-contact features. On the other hand, the standing thermometer requires a large space for installation and the person need to move to the front of the device. In this study, we propose a non-contact thermometer that identifies individuals by acquiring images of wrinkles from a camera module and records body temperature individually. The wrinkle image is cropped and then gamma correction removal and grayscaling are applied. The images are fed into a convolutional neural network for the identification. The experiment using images obtained with a smartphone camera from 12 subjects confirmed that individuals could be identified with an average F-value of 0.94. The experiment using images obtained with the proposed device from five subjects confirmed that the identification accuracy exceeded F-value of 0.93 for all subjects.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

handheld thermometer, body temperature, wrinkle, user identification, neural networks

ACM Reference Format:

Koki Iguma and Kazuya Murao. 2022. Design and Implementation of a Non-contact Thermometer that Records Body Temperature Individually using Forehead Wrinkle Image. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544793.3560395>

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

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ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3560395>

1 INTRODUCTION

With the outbreak of the novel coronavirus (COVID-19), the opportunities to measure body temperature when using public facilities and for daily health checks have increased significantly. The installation of temperature measurement systems using standing thermometers and handheld devices is progressing due to their non-contact nature. Raymond [6] has developed a system that combines facial recognition and an automatic temperature measurement system using a standing thermometer. A standing thermometer requires an installation space, and people need to move to the front of the device. To the best of our knowledge, there are no handheld thermometers that measure body temperatures and link them to individuals. In 2020, in Atlanta, a man lied to his employer about contracting a coronavirus, causing the facility where he worked to temporarily close, resulting in approximately \$100,000 damages [4]. Thus, false reporting of coronavirus infection or body temperature may affect social life, and it must be prevented so that people other than the person in question cannot impersonate the person taking the temperature or falsely report the body temperature.

This paper propose a non-contact thermometer that identifies individuals by acquiring images of wrinkles from a camera module and records body temperature individually. Wrinkles appear on the forehead at the intended timing by contracting the frontal muscles, and images of the wrinkles are acquired from a camera module attached to a non-contact handheld thermometer. The wrinkle image is identified with the convolutional neural network (CNN), then the body temperature is recorded individually. The proposed method assumes that the user intentionally reveals a wrinkle to identify the individual when the user needs to take a temperature. The proposed method is more privacy-conscious than the face recognition method used in standing thermometers and making impersonation difficult as forehead wrinkles are used.

2 RELATED WORK

This section introduces biometric authentication using images, object recognition using convolutional neural networks, research on forehead wrinkles, and research on body temperature and personal identification.

2.1 Biometric Authentication using Images

Wang et al [11] achieved state-of-the-art accuracy on an existing face image dataset by using a new loss function LMCL for face recognition. Yan et al. [12] reported that they obtained 97.8% accuracy in a discrimination scenario based on 3D images of the ear. Park et al. [5] developed a system that automatically analyzes

the quality of urine and stool in toilets for long-term monitoring of human health, and identifies individuals based on fingerprints and anal wrinkles images and associates them with the quality of their wrinkles and stool. Ahmad et al. [1] developed a system to authenticate an individual with less computation and less memory, while taking privacy into consideration, by applying wave atom transformation to palm vein features obtained from a near-infrared camera.

Since biometric information is always visible on the surface of the human body, there is a possibility that it can be stolen in social platforms. Some methods require special cameras. Furthermore, there is a concern about privacy in face recognition, since people can guess who they are just by looking at the face images used. In contrast, the proposed method reveals forehead wrinkle only when the people intend to be reveal it. Since only the forehead image is acquired, the proposed method is more privacy-conscious than a full face image.

2.2 Research on Forehead Wrinkles

Sathik et al. [3] found that facial expressions are important for understanding students' emotions and comprehension, and that forehead wrinkles are useful for identifying facial expressions. They then developed a method to detect the forehead, extract wrinkles, recognize emotions, and attempted to classify whether or not a person in a photograph is expressing emotion. Yap et al. [13] pointed out that when using a fill algorithm to hide blemishes and freckles on the face in a photo, the target of the fill is the entire face, which may make the photo unnatural after the fill. They then used an automatic wrinkle detection algorithm and three different fill algorithms to fill in wrinkles on the face, including forehead wrinkles and evaluated whether the results were natural compared to the original photographs.

Several methods have been proposed to estimate facial expressions from wrinkles on the forehead and to detect and remove wrinkles. However, to the best of the author's knowledge, there are no studies that use forehead wrinkles to identify individuals.

2.3 Research on Body Temperature

Ukai et al. [10] used a thermo-vision camera unaffected by illumination changes to extract regions of the surface temperature of the face that was unaffected by outside temperature, and concluded that a 90% authentication rate was achieved for personal identification. On the other hand, since this method uses the temperature distribution of the face, biometric information that can identify an individual, such as the shape of the nose, can be obtained from the contour of the face. In this paper, we propose a system that identifies individuals using forehead wrinkles to prevent impersonation and to protect privacy and links this to body temperature obtained from a non-contact thermometer.

3 PROPOSED METHOD

Wrinkles are revealed on the forehead by contracting the frontal muscles. The images are acquired with a camera module mounted on a handheld thermometer. The acquired image is then preprocessed and fed into a convolutional neural network to identify the individual.

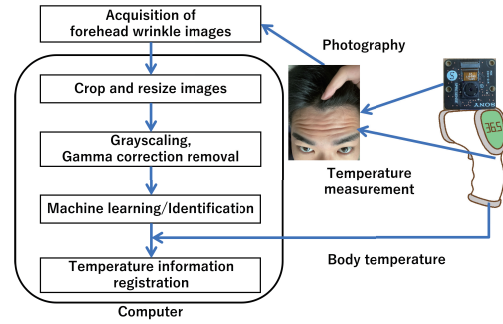


Figure 1: Flow of the proposed method.

3.1 Assumed Environment

A camera module for board computers is used to acquire images of wrinkles on the forehead for personal identification. Since forehead wrinkles do not usually appear on the surface of the body, there is less risk of biometric data theft than with face or fingerprint recognition. Since only the forehead wrinkle is used as biometric information, the system is more privacy-conscious than face recognition. The system requires the user to raise his/her hair with one hand to reveal the forehead wrinkles. Therefore, authentication can be conducted only when they take their temperature using a non-contact thermometer by lifting their hair with one hand. The system is expected to be difficult to disguise and to enable efficient physical condition management.

3.2 Overview of the Proposed Method

Figure 1 shows the flow of the proposed method. An image including forehead wrinkles is acquired from a camera module attached to a handheld thermometer while the temperature is taken. Then, the parts of the image that show forehead wrinkles are cropped, and the images are resized to the input size of the classifier. Then, gamma correction removal and grayscaleing are performed, and the image is fed into a convolutional neural network that has learned people who are identified. At last, the system records the body temperature obtained from the thermometer with individual information such as ID or name.

3.3 Acquisition of Temperature and Forehead Wrinkle Image

The user lifts his/her hair with one hand and makes a wrinkling motion on his/her forehead by opening her eyes wide. A pressure sensor is installed between the button of the handheld thermometer and a switch on a circuit board inside the thermometer. The pressure sensor is used as a trigger to acquire images with a camera module attached to the thermometer. The acquired image and temperature are sent to the PC.

3.4 Image Preprocessing

The area of the forehead wrinkle is cropped from the acquired image so that the image of 161×477 [px] including whole forehead wrinkle is obtained by zooming in/out the original image. Currently, though cropping is done manually, automatic crop is our future

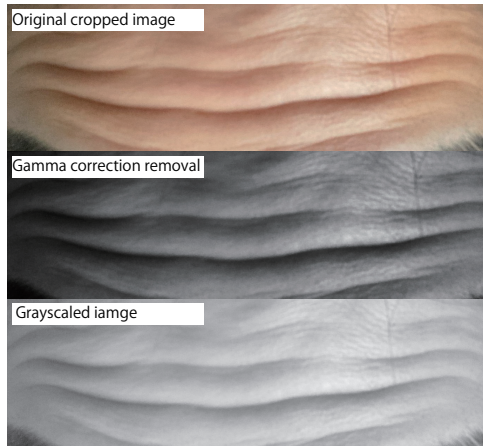


Figure 2: Cropped forehead wrinkle image, image after gamma correction removal, and grayscaled image.

work. A gamma correction removal and grayscaling are applied to the cropped images. Gamma correction is a process in which the Windows OS increases the brightness of an image before displaying it on the screen. This process brightens and corrects the luminance of the middle layer. The difference in luminance between the depressions in the skin and normal skin causes skin wrinkles to appear in the image. The grayscaled image that contains only luminance information facilitates feature extraction by the neural network. Examples of the original image, the image after gamma correction removal, and the grayscaled image are shown in Figure 2.

3.5 User Identification

After the image preprocessing, a convolutional neural network (CNN) is used to learn the forehead wrinkle image. CNN have been applied to image-based classification in various fields: wild chimpanzee face images [2], human gait model [7], handwritten Japanese character [8], pathological diagnosis [9, 15], and tire defect detection [14]. The Pytorch library is used. The structure of the network used in this study is shown in Table 1. Including the input and output layers, the network consists of 10 layers. In Table 1, conv stands for the convolution layer, pool for the pooling layer, bn for the batch normalization layer, and fc for the full-connected layer. The input layer receives image data of $161 \times 477 \times 3$ for color images and $161 \times 477 \times 1$ for grayscaled images. Then, in the convolution layer, the values of each kernel are multiplied by the corresponding image pixel values using a 3×3 kernel (stride=1), and the sum of these values is obtained. In the neural networks used in this study, the first convolution is followed by processing in the batch normalization layer. The batch normalization layer adjusts the mean to 0 and the variance to 1 for each channel of data. The batch normalization layer is introduced to reduce the risk of gradient loss that may occur during training. After three alternating iterations of the convolution and pooling layers, the full concatenation layer converts the matrix data to vector data. The output layer then classifies the data into N people. ReLu is used as the activation function between

Table 1: Structure of the convolutional neural network.

	Length×Width×Channel	Filter
Input	$161 \times 477 \times (1 \text{ or } 3)$	-
conv1	$159 \times 475 \times 16$	3×3
bn1	$159 \times 475 \times 16$	-
pool1	$79 \times 237 \times 16$	2×2
conv2	$77 \times 235 \times 32$	3×3
pool2	$38 \times 117 \times 32$	2×2
conv3	$36 \times 115 \times 64$	3×3
pool3	$18 \times 57 \times 64$	2×2
fc1	$1 \times 1 \times 120$	-
Output	$1 \times 1 \times N$	-

the layers. The parameters are updated using stochastic gradient descent with a learning rate of 0.01.

3.6 Implementation

A non-contact thermometer JPD-FR409-C by Chainabi, Inc. was used. A camera module and Spresense¹ were used to acquire images. A pressure sensor was installed between the button of the thermometer and the switch on the board inside the thermometer. When the button of the thermometer is pressed, the camera module takes a picture of the forehead wrinkle. Temperature data and image are stored on the SD card in the Spresense. The Spresense is connected to the PC via USB-C cable which provides power and data transfer. The PC automatically process the image by monitoring the SD card.

4 EVALUATION EXPERIMENT

To investigate whether forehead wrinkles are effective biometric information for personal identification, we conducted two experiments: experiment 1 to evaluate the accuracy of personal identification by acquiring forehead wrinkle images using a common camera, and experiment 2 to evaluate the accuracy of personal identification using forehead wrinkle images acquired with the implemented device.

4.1 Experiment 1

4.1.1 Experiment Environment. The following procedure was used to collect the forehead wrinkle images from 12 subjects (A–L, in their 20s to 40s). The subjects lifted their forehead hair with one hand and made a movement to raise their eyes widely, which caused their frontal muscles to contract. The wrinkles were photographed with the camera of the iPhoneSE2. Fifty images were taken for each subject, 600 images in total. The wrinkles were made two or three times at intervals, and the pictures were taken. The camera position was slightly shifted up, down, left, and right while taking pictures from various angles. Five-segment cross-validation was used to evaluate accuracy. Cross-validation was repeated five times, and the average values of the precision, recall, and F values were calculated.

¹<https://developer.sony.com/ja/develop/spresense/specifications>

Table 2: Identification accuracy of forehead wrinkle images acquired using a smartphone camera (P: precision, R: recall, F: F-value).

Subject	Original image			Grayscale			Gamma and grayscale		
	P	R	F	P	R	F	P	R	F
A	1.00	0.94	0.97	1.00	1.00	1.00	1.00	0.94	0.97
B	0.92	0.98	0.95	1.00	0.94	0.97	0.94	0.96	0.95
C	0.93	0.78	0.85	0.92	0.68	0.78	1.00	0.86	0.92
D	0.91	1.00	0.95	0.90	0.88	0.89	0.88	0.98	0.92
E	0.92	0.88	0.90	1.00	0.84	0.91	0.93	0.86	0.90
F	0.94	0.98	0.96	1.00	1.00	1.00	1.00	0.98	0.99
G	0.77	0.92	0.84	0.94	0.90	0.92	0.88	0.92	0.90
H	0.84	0.86	0.85	0.79	0.96	0.86	0.89	0.96	0.92
I	0.97	0.68	0.80	0.98	0.96	0.97	0.96	0.98	0.97
J	0.95	0.70	0.80	0.89	1.00	0.94	0.98	1.00	0.99
K	0.64	0.78	0.70	0.82	0.94	0.88	0.91	0.82	0.86
L	0.83	0.98	0.90	0.93	1.00	0.96	0.89	0.98	0.93
Ave.	0.88	0.87	0.87	0.93	0.92	0.92	0.94	0.94	0.94

Table 3: Identification accuracy of forehead wrinkle images acquired using the implemented device (P: precision, R: recall, F: F-value).

Subject	Original image			Grayscale			Gamma and grayscale		
	P	R	F	P	R	F	P	R	F
A	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
B	1.00	1.00	1.00	1.00	0.93	0.97	0.93	0.93	0.93
C	0.97	0.97	0.97	0.97	1.00	0.98	0.94	0.97	0.95
D	0.97	0.93	0.95	0.97	0.93	0.95	1.00	0.80	0.89
E	0.94	0.97	0.95	0.88	0.93	0.95	0.80	0.93	0.86
Ave.	0.97	0.97	0.97	0.96	0.96	0.96	0.93	0.93	0.93

4.1.2 Results and Discussion. Table 2 shows the identification results for the three types of images: original color images, grayscaled images, and images after gamma correction removal and grayscaling. The precision, recall, and F-value of each subject was the highest for the images that combined the gamma correction removal and grayscaling. Among the three types of images, the grayscaling resulted in an average F-value of 0.92, while the combination of gamma correction removal and the grayscaling resulted in an average F-value of 0.94, which was 0.02 higher than the grayscaling only. The original color image had an average F-value of 0.87, which was 0.05 higher for the grayscaling. Therefore, we believe that the pre-processing of grayscaling and gamma correction removal enhances identification accuracy.

4.2 Experiment 2

4.2.1 Experiment Environment. Five subjects A–E same as in experiment 1 joined the experiment 2. Thirty images of forehead wrinkle are taken for each subject using the implemented device, for a total of 150 images. The subject made a wrinkle movement two or three times, and several pictures were taken per one wrinkle. The camera was positioned in the front, left, and right directions.

4.2.2 Results and Discussion. The classification results of the wrinkle images acquired using the implemented device are shown in Table 3. Although the number of classification labels was smaller

than Experiment 1, an average F-value of over 0.93 was obtained for all five subjects for all image types with a general camera module. Therefore, we consider that there is a possibility that the camera module can be used to identify individuals with high accuracy. In the future, we will increase the number of people to be identified and investigate the effect on identification accuracy. In addition, the number of images used in the training data will be reduced to as few as possible by using data expansion.

5 CONCLUSION

In this paper, we proposed a non-contact handheld thermometer in which a camera module is mounted to perform user identification with forehead wrinkle image at the time of measuring the temperature. The proposed method implements two image conversions: gamma correction removal and grayscaling. In the first experiment, forehead wrinkle images were captured and classified using a smartphone, and a high average F-value of 0.94 was obtained. On the other hand, the number of labels classified in this experiment was twelve (twelve persons). In addition, the results of the classification of wrinkle images captured by the proposed device showed that the identification accuracy exceeded 0.93 for each image typespreprocessing pattern, indicating that even a camera module can identify individuals using forehead wrinkles.

For image preprocessing, the wrinkles in the acquired images were manually cropped in this study. The pictures taken by the proposed device include parts of the face other than the wrinkles, such as the eyes and nose. To identify individuals at the same time as temperature measurement, we will develop a system that automatically crops the area of the wrinkles. We also consider revising the network structure because there is a possibility that the identification rate will decrease as the number of persons to be identified increases. We will reduce the number of wrinkle images for training as much as possible by applying data augmentation.

ACKNOWLEDGMENTS

This research was supported in part by the Japan Science and Technology Agency, PRESTO grant number JPMJPR1937, Japan.

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